Project Initialization and Planning Phase

Date	27 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5Marks

Project Proposal (Proposed Solution) Report:

Project Overview	
Objective	The primary objective is to revolutionize the loan approval process by implementing advanced machine learning techniques, ensuring faster and more accurate assessments.
Scope	The project comprehensively assesses and enhances the loan approval process, incorporating machine learning for a more robust and efficient system.

Problem Statement	
Description	Addressing inaccuracies and inefficiencies in the current loan approval system adversely affects operational efficiency and customer satisfaction.
Impact	Solving these issues will result in improved operational efficiency, reduced risks, and an overall enhancement in the lending process, contributing to customer satisfaction and organizational success.

Proposed Solution	
Approach	Employing machine learning techniques to analyze and predict creditworthiness, creating a dynamic and adaptable loan approval system.
Key Features	• Implementation of a machine learning-based credit assessment model. • Real-time decision-making for quicker loan approvals. • Continuous learning to adapt to evolving financial landscapes. • Automated data processing and feature engineering. • Model explainability for transparency and trust. • Dashboard for performance monitoring and insights.

Resource Requirements		
Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	1 Server with NVIDIA T4 GPU, 8 vCPU (16 cores)
Memory	RAM specifications	32 GB RAM
Storage	Disk space for data, models, and logs	2 TB SSD
Software		
Frameworks	Python frameworks	Scikit-learn, Pandas, NumPy, Flask
Libraries	Additional libraries	Matplotlib, Seaborn, XGBoost, Imbalanced-learn, Joblib
Development Environment	IDE	Jupyter Notebook, PyCharm
Version Control	Code repository	Git, GitHub
Data		
Data	Source, size, format	Kaggle Loan Prediction Dataset (614 records, CSV), UCI German Credit Dataset (1000 records, CSV), Internal historical loan data (if available)